ANESU RIRWA

AI & Machine Learning Engineer
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PROFESSIONAL SUMMARY

Data & AI Engineer and Technology Lead at Munisa Resources, a Zimbabwean resources and infrastructure group, promoted four months after joining. Strong in data analytics and business intelligence: live dashboards, board-level reporting and analysis that uncovered confirmed gold and diesel theft. Builds the systems organisations run on, from an operations and finance platform to a production forecasting model. Also runs Kordel Data for mining and healthcare clients. BSc (Hons) AI and Machine Learning (2:1); seeking data, AI or analytics roles.

PROFESSIONAL EXPERIENCE

Munisa Resources Group — Harare, Zimbabwe

Resources and infrastructure group with subsidiaries in mining, quarrying and civil works, and commodity trading.

Data & AI Engineer and Technology Lead — November 2025 – Present

- Built a Python machine learning model forecasting production grade, tonnage and costs (R^2 = 0.82), now used for budgeting, including for heavy equipment.
- Lead the group's technology roadmap across the parent company and its subsidiaries, reporting to the board and working closely with the Group Managing Director and Executive Director.
- Designed and built **MRIS**, the group's operations and finance system (Next.js, TypeScript, PostgreSQL), used by 15 staff across subsidiaries; delivered in phases so reporting never stopped.
- Built and launched **DealRoom** (dealroom.munisaresources.com), a marketplace connecting mineral claim holders with investors (18 claims listed with 13 more being added, 4 deals in progress).
- Lead group data analysis and business intelligence, producing weekly, monthly and board-level reports; redesigned the group's three corporate websites.

Data Analyst — July 2025 – October 2025

- Centralised records and built live Power BI dashboards, replacing separate spreadsheets and paper logbooks; cut manual data preparation by 70% and ended disputes over figures.
- Uncovered underground gold theft by comparing expected and actual milling grades (shortfall rose from 15% to 67% over three runs before being confirmed); dashboards also exposed 200 litres of diesel theft.

Kordel Data · kordeldata.com — Harare, Zimbabwe

Founder and Principal Consultant (alongside full-time role) — January 2026 – Present

- **Maphompe Minerals:** designed a structured data capture system for 10 staff, standardising operational data and enabling live dashboards.
- **Capital Physiotherapy:** building the clinic's website and patient records system, designed for Zimbabwe's Cyber and Data Protection Act.
- **Products:** built and launched **ImariiMaCharges** (imariimacharges.kordeldata.com), which calculates EcoCash transaction charges from real transaction data, and **Stridetale** (stridetale.kordeldata.com, in beta), a running and lifting tracker that turns training into a journey to a real destination.

Gifted Community Organisation — Harare, Zimbabwe

Community organisation funding school tuition and providing clothing for children in Macheke.

Data Analyst & Software Engineer — January 2025 – June 2025

- Built an Excel and Python model ranking pupils on academic and financial records (in 2025, the 10 most in need each term were shortlisted and all funded), and the organisation's website.

Webgems Harare, Zimbabwe

Software company building websites, mobile apps and business systems for clients.

Software Engineer Intern (12-month industrial attachment) August 2022 – August 2023

- Built client websites (WordPress) and cross-platform mobile apps (Ionic with Angular and React); designed and implemented a Fiscal Invoice Generator. Attachment graded First Class by the University of Zimbabwe.

SELECTED PROJECTS

- **Shona Next-Word Prediction (NLP):** built a next-word prediction model for Shona from a corpus I extracted from PDF documents; trained Word2Vec embeddings and compared two bidirectional LSTM models (TensorFlow/Keras).
- **FIFA World Cup 2026 Match Prediction:** XGBoost models trained on 7,900 international matches; removed a data leak and used time-based validation and probability calibration, reaching 59.9% result accuracy against a 47.5% baseline, benchmarked against bookmaker odds.

EDUCATION

University of Zimbabwe August 2020 – July 2024

BSc (Honours) Artificial Intelligence and Machine Learning · Upper Second Class Honours (2:1)

- **Final-year project:** Optimising customer service operations for SMEs and solo founders using AI and machine learning, including a Shona–English natural language processing component.
- **Selected modules:** Machine Learning and Neural Networks, Deep Learning, NLP, Big Data Technologies, Data Structures and Algorithms (First Class).

SKILLS

Proficient: Python (Pandas, NumPy), SQL, data cleaning and preparation, data visualisation, PostgreSQL and Supabase, relational database design, Power BI, Excel, SharePoint, TypeScript, Next.js, Git, WordPress

Working knowledge: scikit-learn, XGBoost, Random Forest, predictive modelling (regression and forecasting), NLP, TensorFlow, PyTorch, React, Flask, ETL pipelines, Tableau, Ionic

CERTIFICATIONS AND TRAINING

- Machine Learning Specialization — Andrew Ng, DeepLearning.AI and Stanford Online, via Coursera (in progress)
- Python 101 for Data Science (PY0101EN) — IBM, via Cognitive Class (May 2025)
- All In Africa Program — All In Open Source: open-source development curriculum, career readiness workshops and hackathons (November 2023 – February 2024)
- IT Essentials — Cisco Networking Academy, University of Zimbabwe (July 2021)

LEADERSHIP AND VOLUNTEERING

- **Uncommon.org — Volunteer Mentor (November 2023 – May 2025):** mentored three students weekly for six months each in programming, data science, CVs and interviews; all three secured internships and progressed to full-time roles, and one returned as an instructor.
- **Gifted Community Organisation — Volunteer (June 2025 – Present):** maintain the website, organise clothing donations and support the tuition programme.
- **University of Zimbabwe — Elected Class Representative (2020–2024)**

LANGUAGES

- **English** — C1 (Duolingo English Test 140, October 2026)
- **Shona** — Native
- **German** — A2 (A1 certified; studying B1)